

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

Total Marks: 100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I T.V.S. Jagadeesh ¹¹⁹²⁴⁷²¹⁴⁶ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

T. S. Jagadeesh
Signature of the student:

Answer to all :

1. A college wants to identify students who may fail in the final exam. How can Data Science help?
2. A hospital has patient records for many years. How can Data Science be applied?
3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?
4. A bank wants to detect suspicious transactions. How can Data Science be useful?
5. A company wants to reduce customer loss. How can Data Science support this?
6. A school wants to improve student performance. Mention one benefit of Data Science in this case.
7. A supermarket wants to know which products sell more during festivals. How can Data Science help?
8. A transport department wants to reduce traffic congestion. How can Data Science be applied?
9. A farmer wants to improve crop yield. Give one use of Data Science.
10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?
11. A weather app updates temperature every minute. Which facet of data is involved?
12. A university stores student details in rows and columns. What type of data structure is this?
13. Customer comments on social media are collected for analysis. What type of data is this?
14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?
15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?
16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?
17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?
18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?
19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?
20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Organization	10	Clear, concise, and well-organized	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Criteria	Weightage	No. of Questions	Marks Obtained
Conceptual Understanding	30	6	20
Accuracy of Answer	25	5	20
Application of Knowledge	20	4	20
Completeness of Response	15	3	15
Clarity & Presentation	10	2	10
TOTAL			95

95
100

C01-AT1

- ① To identify student college data:

Data science uses predictive analytics to analyze past students data such as marks, attendance, and assignments. Machine learning model can identify patterns of weak students. This helps teachers take early action like extra coaching. As a result, failure rates can be reduced.

- ② Hospital has patient records for many years.

Data science can analyze historical patient data to find patterns in diseases and treatments. It helps in predicting future health risks and improving diagnosis. Hospitals can optimize resource and provide better care.

- ③ An online shopping website wants to recommend product to customers.

This is a Recommendation system. It uses customer data like purchase history and browsing behavior. Algorithms suggest relevant products based on user preference. This improves user experience and increases sales.

- ④ A bank wants to detect suspicious transaction.

Data science helps in fraud detection by analyzing transaction patterns. Machine learning models detect unusual or abnormal activities.

- ⑤ A company wants to reduce customer loss.

Data science uses customer churn analysis to identify customers who may leave. It analyzes usage patterns, complaints, and behaviour. Companies can take preventive action like offers or support.

⑥ A school wants to improve student performance.
Data science enables personalized learning.
It analyze student performance and learning styles. Teachers can give targeted guidance and support. This helps improve academic.

⑦ A supermarket wants to know which products.
Data science analyze past sales data identify seasonal trends. It shows which products have high demand during festival. This helps in inventory planning and promote.

⑧ A transport department wants to reduce.
Data science uses data from GPS and to analyze traffic patterns. It helps in prediction and route optimization. signals can be managed efficiently. reduce congestion and travel time.

⑨ A farmer wants to improve crop yield.
Data science helps in crop prediction and soil analysis. It uses weather, soil and past yield data. Farmers can select suitable crops and improving productivity. This increase crop yield.

⑩ A company has data in Excel tables, customer reviews, images, and videos.

This represents variety. Data is available in different formats such as structured (tables) and unstructured (text, images, videos) multiple data types is a big challenge at

⑪ A weather app updates temperature and velocity. Data is generated and processed continuously at high speed. Real-time updates require fast data processing systems.

⑫ A university stores student details in rows and columns.

This is structured data. It is organized in a tabular format with row and columns. It is easy to store, query, and analyze using databases.

⑬ Customer comments on social media are collected for analysis.

This is Unstructured data. It does not follow a fixed format. It includes text, images and videos. Special technique like NLP are used for analysis.

⑭ A company collects a very large amount of daily data.

This represents volume. A huge amount of data is generated daily. Big data tools are required to store and process such large datasets.

⑮ A hospital dataset contain missing age values and incorrect phone numbers.

This shows data quality issue such as missing values and incorrect data. These problems affect the accuracy of analysis. Data cleaning is necessary before processing.

(16) A business wants to convert customer data into useful sales decisions.

This represents value. The main goal of this data is to extract meaningful insights. These insights help businesses make better decisions.

(17) A company stores large files in the cloud.

This uses Hadoop (HDFS), which is a distributed storage system, it stores large data across multiple machines. It ensures scalability and reliability.

(18) A company wants to process large datasets using parallel computation.

Apache Spark is used for parallel data processing. It processes large datasets quickly and efficiently. It is faster than traditional methods.

(19) An organization wants to collect live sensor data from IoT devices.

Apache Kafka is used for real-time data streaming. It collects and processes live sensor data efficiently. It ensures fast and reliable data transfers.

(20) A company wants to visualize big data results for decision makers.

Data visualization tools like Tableau, Power BI or matplotlib are used. They present data using charts and graphs. This helps decision makers understand insight easily.